

What are Comprehensions?

Comprehensions are concise, readable way to create list, set & dictionary by iterating over a sequence & optionally applying a condition.

- all in a single line of code.

① Create easily - list, set, dict

② Iterate & assign result to container data type
- list, set, dict

③ Not for tuple - tuples are immutable.

] List Comprehension - List Comprehension creates a new list by iterating over a sequence &

applying an expression to each item/element.

— optionally filtering with condition.

Syntax :-

[expression for item in
iterable(sequence if condition)]

logic

1. list
2. set
3. dictionary

optional

$\boxed{\text{item} > 5}$

Q. Create a list having square of no. from 1 to 5

exp o/p - $\left[\underset{1}{1}, \underset{2}{4}, \underset{3}{9}, \underset{4}{16}, \underset{5}{25} \right]$

$$\begin{aligned} 1^2 &= 1 \\ 2^2 &= 4 \\ 3^2 &= 9 \\ 4^2 &= 16 \\ 5^2 &= 25 \end{aligned}$$

without Comprehension:-

~~1, 2, 3, 4, 5~~

$\text{square} = (1, 4, 9, 16, 25)$

for i in range(1,6,1):

print(i ** 2)

o/p -
1
4
9
16
25

exponentiation - $2 ** 2 = 2^2$
 $3 ** 2 = 3^2$

list, set, dict
range

with Comprehension -

Syntax: [expression for item in iterable
if condition]

list1 = [i ** 2 for i in range(1,6,1)]

print(list1) → [1, 4, 9, 16, 25]